

ASMI BASNET

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SUMMARY

Applied AI engineer (M.S. Analytics) who builds RAG systems, tool-calling agents and ML models in Python. SQL and data-modeling background, so the systems run on real, validated data.

EXPERIENCE

- Graduate Researcher, SPS Corps - Analytics Initiative**

Saint Louis University | Aug 2025 - Present

 - Built a RAG decision assistant (local Ollama models plus OpenAI API, vector index) so non-technical staff can query institutional databases and reports in plain English instead of requesting custom SQL.
 - Built a hybrid forecasting engine (Prophet plus LSTM) with automated anomaly detection, giving leadership budget and demand forecasts with flagged outliers each quarter.
 - Wrote automated data-quality checks (referential integrity, schema drift detection) that stop bad records from reaching the assistant and reports, in line with FERPA privacy rules.
 - Wrote SQL (CTEs, window functions) across 8+ operational tables as the data layer for the assistant and reports, and documented a data dictionary.
- Data Analyst Intern / BI Developer**

Saint Louis University | Jun 2024 - Aug 2024

 - Built 6 Power BI and Tableau dashboards on star-schema models, tracking 15+ institutional KPIs for university leadership.
 - Added automated validation and audit checks on sensitive student records to meet FERPA and governance requirements.
- Data Analyst / Researcher & Systems Analyst**

Entegra Sources, Nepal | Dec 2022 - Jul 2023

 - Cleaned, transformed and mapped survey, transactional and operational data into standardized relational schemas using Python (Pandas, SQLAlchemy) and SQL.
 - Analyzed 5,000+ survey responses in SPSS and R (regression, ANOVA, chi-square) and wrote methodology and audit documentation.

PROJECTS

- PolicyGuard: Offline RAG Compliance Suite**

Ollama, ChromaDB, FastAPI, Streamlit, Docker

 - Runs fully offline, so policy documents never leave the machine. A verification step checks citations, flags contradictory policies and asks for missing context, so reviewers see conflicts with sources instead of reading whole handbooks.
 - Produces a governance health score from the conflicts found.
- Nepal Disaster Watch (WebMCP Challenge)**

TypeScript, WebMCP, Leaflet, REST APIs

 - Exposes 11 native WebMCP tools (6 read, 5 write) over six disaster data sources, including cross-referencing evidence across all 77 districts and driving the live map in sync with a human responder.
 - Every feed reports its own live, unreachable or snapshot status, so failures are visible. Live demo and code linked on my portfolio.
- CareGap: Clinical Decision Support Prototype**

DuckDB, XGBoost, SHAP, FAISS, Ollama, Docker

 - Risk model (Lasso, Random Forest, XGBoost) on 34,000+ **synthetic** (Synthea) patient records with per-patient SHAP explanations and a what-if simulator.
 - RAG assistant with automatic failover from the local model to Gemini, and guardrails that block dosage suggestions.
- Multi-Agent Financial Research System**

Python, Agno, OpenAI GPT-4o, Zapier

 - Two cooperating agents (Market Scout, Financial Analyst) with tool execution and iterative query refinement turn live market news into structured briefs; Zapier webhooks triage alerts and send summaries.

SKILLS

LLM systems	RAG, embeddings (SentenceTransformers), vector search (FAISS, ChromaDB), Ollama (Llama 3, Qwen 2.5, Phi-3), OpenAI API, Gemini API, Agno, MCP, prompt engineering, guardrails
Machine learning	PyTorch (CNN, ResNet), scikit-learn, XGBoost, LightGBM, SHAP, Prophet, LSTM, cross-validation
Engineering	Python (FastAPI, Streamlit, Pandas, SQLAlchemy), TypeScript, REST APIs, Docker Compose, Git/GitHub, Linux CLI
Data	SQL (PostgreSQL, SQL Server, DuckDB, SQLite, Snowflake, BigQuery), CTEs, window functions, dimensional modeling, ETL

EDUCATION

- M.S. Analytics (STEM)**

Saint Louis University, MO | May 2026
- B.E. Electronics & Communication Engineering (STEM)**

Tribhuvan University, Nepal | 2017 - 2022